

Day 11 — Content Modifier, Part 2 (25 Aug 2026)

1. Recap — What Content Modifier Does

What will Content Modifier do? → It will Add, Modify, Delete the data.

2. Options Inside Content Modifier — General Tab

Options in Content Modifier → General Tab:

- Message Header
- Exchange Property
- Message Body

3. Message Header — Recap, Precisely Worded

If we define any variable in Message Header → it will go to the Target System as a "variable header."

4. Exchange Property — Recap, Precisely Worded

Exchange Property: The variable that is stored in the Exchange Property will be used **inside that iFlow only**.

5. Message Body — Why It's the Most Powerful

Message Body: This is the place where the message is sent to the Target System. It is the **most powerful** of the three, because **whatever you do in the Message Body part, that exact same data will be sent to the Target System**.

Why "most powerful" is the right way to think about this: Unlike a single header or property (which carries one specific piece of data), the Message Body IS the actual payload — the entirety of what the receiving system sees and processes as the "real" message. Any change here directly becomes the receiver's input, which is why developers need to be especially careful and deliberate when touching the Message Body.

6. Recap: Source Types Covered So Far (Constant & Expression)

Yesterday we discussed the Source Types in Message Header, which are:

#	Source Type	Description
1	Constant	A hard-coded value → static value output
2	Expression	<code>\${header.headervalue}</code> or more generally <code>\${...}</code>

7. Today's Focus: Property and Header-via-XPath Source Types

8. XPath — Full Explanation

XPath: It's a dynamic value.

Source	→	Capture	→	Use (XPath)
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Worked Example

```
<orders>
  <id>123</id>
  <status>pending</status>
</orders>
```

XPath Expression for "id"

XPath Style	Expression
Absolute XPath	<code>/orders/order/id</code>
Relative XPath	<code>//id</code>

Verified accurate: A single leading slash (/) starts an **Absolute XPath** — it defines the full, exact path from the document's root element. A double slash (//) is a **Relative XPath** — it searches for a matching element *anywhere* in the document, regardless of its exact position in the structure. This is standard, correct XPath syntax, and the distinction matters in real projects: Absolute XPath is more precise but breaks if the XML structure changes slightly; Relative XPath is more flexible but can accidentally match more than one element if the tag name repeats elsewhere in the document.

9. XPath — Data Type Field

XPath → Data Type: `java.lang.String` (note the capital "S" in String — Java class names are case-sensitive)

10. The Key Question: How Do We Use These Values in the Body?

Question: If variables are created in Message Header or Exchange Property using Constant and XPath (Source Types), then how do we use them in the Body?

Answer: We will use Expression (which is an **Apache Camel expression**):

- `${in.body}`
- `${header.headervalue}`
- `${property.propertyvalue}`

Why this matters: This is the unifying principle tying together everything from Day 10 and Day 11 — no matter where a value originally came from (Constant, XPath, Header, Property), you always pull it into the Message Body the same way: through a Camel Simple Expression wrapped in `${ }`. Once you're comfortable with this single pattern, you can combine values from anywhere in the message into your final payload.

11. Hands-On Demo — Done in Trial Account

iFlow name used for this demo: `Day_11_https_adapter_and_content_modifier`

Summary Table — What Goes Where

Variable Type	What Happens When Created
Header	Variable created → goes to Target Headers (visible to the receiving system)
Property	Variable created, value assigned → stays inside the flow only — No appearance in Target System Headers
Body	→ Sends directly to the Target System (this IS what the receiver gets as its actual payload)

This table is the cleanest summary of the entire Content Modifier topic across Days 10-11 — worth memorizing as-is. If asked in an interview "explain the difference between header, property, and body in CPI," this three-row table is essentially the complete, correct answer.